

✓ 1. Log-likelihood under Fréchet

Recall Fréchet:

$$f(x) = A\alpha(Ax)^{-(1+\alpha)}e^{-(Ax)^{-\alpha}},$$

$$F(x) = e^{-(Ax)^{-\alpha}}$$

Use:

$$\log f(x) = \log(A\alpha) - (1 + \alpha)\log(Ax) - (Ax)^{-\alpha}$$

$$\log F(x) = -(Ax)^{-\alpha}$$

◆ Take logs of your likelihood

$$\begin{aligned} \ell(\gamma; A, \alpha) &= \left(\sum_{n=1}^{N_T} l_n \right) \log \gamma + \sum_{n=1}^{N_T} \log f(x_{l_n}) + \sum_{n=1}^{N_T} (\gamma^{l_n} - 1) \log F(x_{l_n}) \\ &+ \sum_{n=1}^{N_T-1} \frac{\gamma^{l_{n+1}} - \gamma^{l_n}}{1 - \gamma} \log F(x_{l_n}) + \mathbf{1}_{l_{N_T} < T} \frac{\gamma^{l_{N_T}+1} - \gamma^{T+1}}{1 - \gamma} \log F(x_{l_{N_T}}) \end{aligned}$$

✓ Final log-likelihood

$$\begin{aligned} \ell &= \left(\sum l_n \right) \log \gamma + N_T \log(A\alpha) - (1 + \alpha) \sum \log(Ax_{l_n}) \\ &\quad - \sum (Ax_{l_n})^{-\alpha} \\ &\quad - \sum (\gamma^{l_n} - 1) (Ax_{l_n})^{-\alpha} \\ &\quad - \sum_{n=1}^{N_T-1} \frac{\gamma^{l_{n+1}} - \gamma^{l_n}}{1 - \gamma} (Ax_{l_n})^{-\alpha} \\ &\quad - \mathbf{1}_{l_{N_T} < T} \frac{\gamma^{l_{N_T}+1} - \gamma^{T+1}}{1 - \gamma} (Ax_{l_{N_T}})^{-\alpha} \end{aligned}$$

OR:

$$z_n = (Ax_{l_n})^{-\alpha}, \quad \log_n = \log(Ax_{l_n})$$

$$\begin{aligned}
\ell(\gamma, A, \alpha) = & \left(\sum_{n=1}^{N_T} l_n \right) \log \gamma \\
& + N_T \log(A\alpha) \\
& - (1 + \alpha) \sum_{n=1}^{N_T} \log(Ax_{l_n}) \\
& - \sum_{n=1}^{N_T} z_n \\
& - \sum_{n=1}^{N_T} (\gamma^{l_n} - 1) z_n \\
& - \sum_{n=1}^{N_T-1} \frac{\gamma^{l_{n+1}} - \gamma^{l_n}}{1 - \gamma} z_n \\
& - \mathbf{1}_{l_{N_T} < T} \frac{\gamma^{l_{N_T}+1} - \gamma^{T+1}}{1 - \gamma} z_{N_T}
\end{aligned}$$

✓ 4. Compact form (recommended)

Define weights:

$$w_n = 1 + (\gamma^{l_n} - 1) + \frac{\gamma^{l_{n+1}} - \gamma^{l_n}}{1 - \gamma}$$

(and last term adjusted if $n = N_T$).

Then:

$$\ell(\gamma, A, \alpha) = \left(\sum l_n \right) \log \gamma + N_T \log(A\alpha) - (1 + \alpha) \sum \log(Ax_{l_n}) - \sum w_n (Ax_{l_n})^{-\alpha}$$

✓ 2. First-order derivatives

Define:

$$z_n = (Ax_{l_n})^{-\alpha}, \log_n = \log(Ax_{l_n})$$

◆ w.r.t. A

$$\begin{aligned}\ell_A = & -\frac{\alpha N_T}{A} + \alpha \sum x_{l_n} (Ax_{l_n})^{-\alpha-1} \\ & + \alpha \sum (\gamma^{l_n} - 1) x_{l_n} (Ax_{l_n})^{-\alpha-1} \\ & + \alpha \sum c_n x_{l_n} (Ax_{l_n})^{-\alpha-1}\end{aligned}$$

where c_n gathers the two last weights.

◆ w.r.t. α

$$\begin{aligned}\ell_\alpha = & \frac{N_T}{\alpha} - \sum \log_n \\ & + \sum z_n \log_n \\ & + \sum (\gamma^{l_n} - 1) z_n \log_n \\ & + \sum c_n z_n \log_n\end{aligned}$$

◆ w.r.t. γ

Let:

$$S_1 = \sum l_n$$

$$\begin{aligned}\ell_\gamma = & \frac{S_1}{\gamma} + \sum l_n \gamma^{l_n-1} \log F(x_{l_n}) \\ & + \sum \partial_\gamma \left(\frac{\gamma^{l_n+1} - \gamma^{l_{n+1}}}{1 - \gamma} \right) \log F(x_{l_n}) \\ & + \text{last term}\end{aligned}$$

(derivative of rational term handled in code).



3. Second-order derivatives (structure)

All follow from:

- $\partial_A^2 (Ax)^{-\alpha} = \alpha(\alpha + 1)x^2(Ax)^{-\alpha-2}$
- $\partial_\alpha^2 = -(Ax)^{-\alpha}(\log(Ax))^2$

◆ Main Hessian elements

AA

$$\ell_{AA} = \frac{\alpha N_T}{A^2} - \alpha(\alpha + 1) \sum w_n x_{l_n}^2 (Ax_{l_n})^{-\alpha-2}$$

$\alpha\alpha$

$$\ell_{\alpha\alpha} = -\frac{N_T}{\alpha^2} - \sum w_n z_n (\log_n)^2$$

$A\alpha$

$$\ell_{A\alpha} = -\frac{N_T}{A} + \sum w_n x_{l_n} (Ax_{l_n})^{-\alpha-1} (1 - \alpha \log_n)$$

Terms with γ

They involve:

$$\frac{d}{d\gamma} \gamma^k = k\gamma^{k-1}, \quad \frac{d}{d\gamma} \frac{\gamma^a - \gamma^b}{1 - \gamma}$$

(computed numerically in R for robustness).